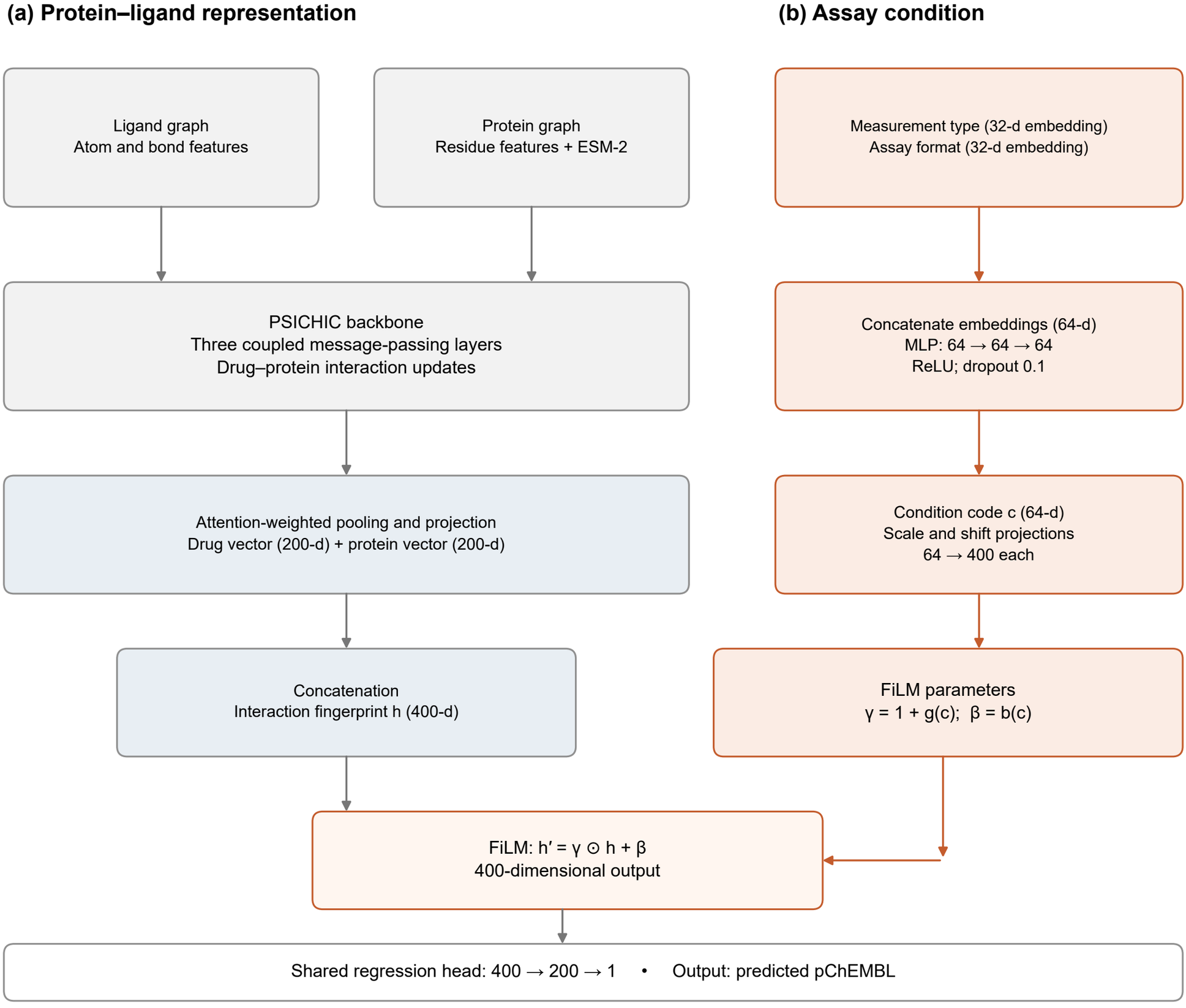
(a) Protein–ligand representation
(b) Assay condition
Ligand graph
Atom and bond features
Protein graph
Residue features + ESM-2
Measurement type (32-d embedding)
Assay format (32-d embedding)
PSICHIC backbone
Three coupled message-passing layers
Drug–protein interaction updates
Concatenate embeddings (64-d)
MLP: 64 → 64 → 64
ReLU; dropout 0.1
Attention-weighted pooling and projection
Drug vector (200-d) + protein vector (200-d)
Condition code c (64-d)
Scale and shift projections
64 → 400 each
Concatenation
Interaction fingerprint h (400-d)
FiLM parameters
γ = 1 + g(c); β = b(c)
FiLM: h′ = γ ⊙ h + β
400-dimensional output
Shared regression head: 400 → 200 → 1 • Output: predicted pChEMBL